

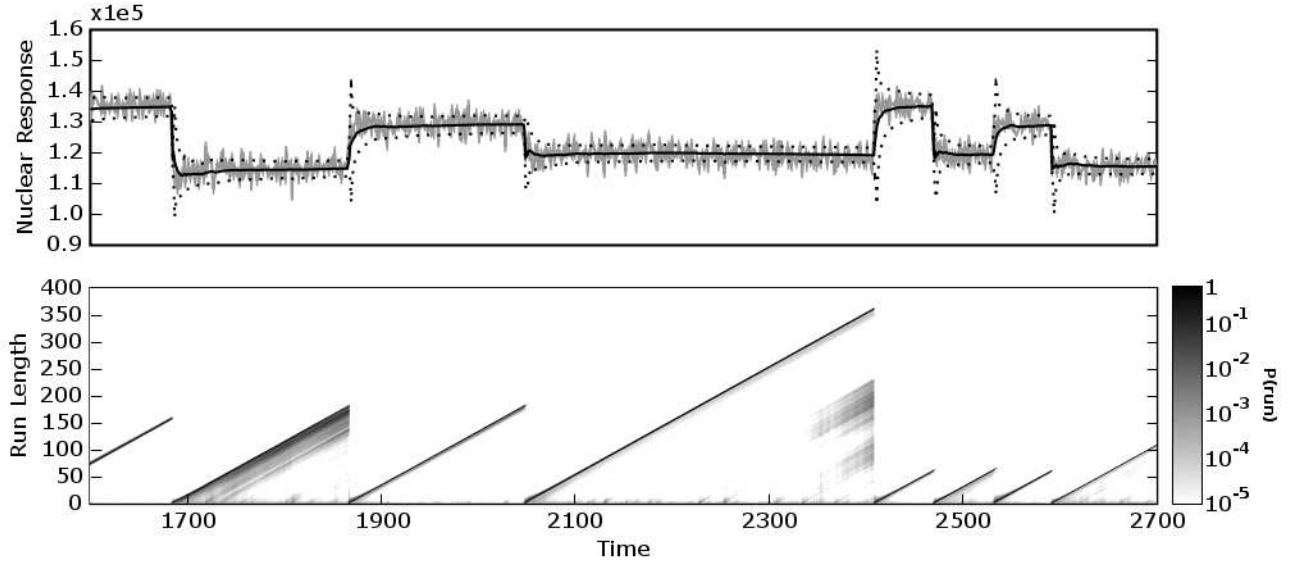


Figure 2: The top plot is a 1100-datum subset of nuclear magnetic response during the drilling of a well. The data are plotted in light gray, with the predictive mean (solid dark line) and predictive $1\text{-}\sigma$ error bars (dotted lines) overlaid. The bottom plot shows the posterior probability of the current run $P(r_t | \mathbf{x}_{1:t})$ at each time step, using a logarithmic color scale. Darker pixels indicate higher probability.

2.4 COMPUTATIONAL COST

The complete algorithm, assuming exponential-family likelihoods, is shown in Algorithm 1. The space- and time-complexity per time-step are linear in the number of data points so far observed. A trivial modification of the algorithm is to discard the run length probability estimates in the tail of the distribution which have a total mass less than some threshold, say 10^{-4} . This yields a constant average complexity per iteration on the order of the expected run length $E[r]$, although the worst-case complexity is still linear in the data.

3 EXPERIMENTAL RESULTS

In this section we demonstrate several implementations of the changepoint algorithm developed in this paper. We examine three real-world example datasets. The first case is a varying Gaussian mean from well-log data. In the second example we consider abrupt changes of variance in daily returns of the Dow Jones Industrial Average. The final data are the intervals between coal mining disasters, which we model as a Poisson process. In each of the three examples, we use a discrete exponential prior over the interval between changepoints.

3.1 WELL-LOG DATA

These data are 4050 measurements of nuclear magnetic response taken during the drilling of a well. The data are used to interpret the geophysical structure of the

rock surrounding the well. The variations in mean reflect the stratification of the earth’s crust. These data have been studied in the context of changepoint detection by Ó Ruanaidh and Fitzgerald [21], and by Fearnhead and Clifford [12].

The changepoint detection algorithm was run on these data using a univariate Gaussian model with prior parameters $\mu = 1.15 \times 10^5$ and $\sigma = 1 \times 10^4$. The rate of the discrete exponential prior, λ_{gap} , was 250. A subset of the data is shown in Figure 2, with the predictive mean and standard deviation overlaid on the top plot. The bottom plot shows the log probability over the current run length at each time step. Notice that the drops to zero run-length correspond well with the abrupt changes in the mean of the data. Immediately after a changepoint, the predictive variance increases, as would be expected for a sudden reduction in data.

3.2 1972-75 DOW JONES RETURNS

During the three year period from the middle of 1972 to the middle of 1975, several major events occurred that had potential macroeconomic effects. Significant among these are the Watergate affair and the OPEC oil embargo. We applied the changepoint detection algorithm described here to daily returns of the Dow Jones Industrial Average from July 3, 1972 to June 30, 1975. We modelled the returns

$$R_t = \frac{p_t^{\text{close}}}{p_{t-1}^{\text{close}}} - 1, \quad (14)$$

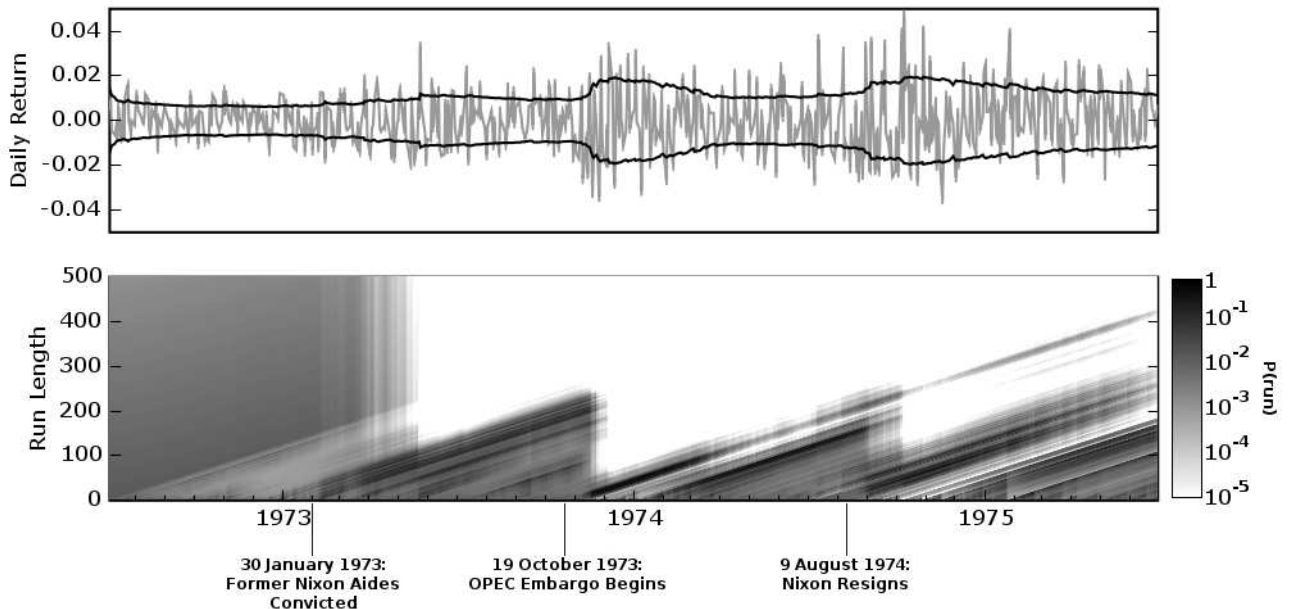


Figure 3: The top plot shows daily returns on the Dow Jones Industrial Average, with an overlaid plot of the predictive volatility. The bottom plot shows the posterior probability of the current run length $P(r_t | \mathbf{x}_{1:t})$ at each time step, using a logarithmic color scale. Darker pixels indicate higher probability. The time axis is in business days, as this is market data. Three events are marked: the conviction of G. Gordon Liddy and James W. McCord, Jr. on January 30, 1973; the beginning of the OPEC embargo against the United States on October 19, 1973; and the resignation of President Nixon on August 9, 1974.

(where p^{close} is the daily closing price) with a zero-mean Gaussian distribution and piecewise-constant variance. Hsu [14] performed a similar analysis on a subset of these data, using frequentist techniques and weekly returns.

We used a gamma prior on the inverse variance, with $a = 1$ and $b = 10^{-4}$. The exponential prior on changepoint interval had rate $\lambda_{\text{gap}} = 250$. In Figure 3, the top plot shows the daily returns with the predictive standard deviation overlaid. The bottom plot shows the posterior probability of the current run length, $P(r_t | \mathbf{x}_{1:t})$. Three events are marked on the plot: the conviction of Nixon re-election officials G. Gordon Liddy and James W. McCord, Jr., the beginning of the oil embargo against the United States by the Organization of Petroleum Exporting Countries (OPEC), and the resignation of President Nixon.

3.3 COAL MINE DISASTER DATA

These data from Jarrett [15] are dates of coal mining explosions that killed ten or more men between March 15, 1851 and March 22, 1962. We modelled the data as an Poisson process by weeks, with a gamma prior on the rate with $a = b = 1$. The rate of the exponential prior on the changepoint interval was $\lambda_{\text{gap}} = 1000$. The data are shown in Figure 4. The top plot shows the cumulative number of accidents. The rate of the

Poisson process determines the local average of the slope. The posterior probability of the current run length is shown in the bottom plot. The introduction of the Coal Mines Regulations Act in 1887 (corresponding to weeks 1868 to 1920) is also marked.

4 DISCUSSION

This paper contributes a predictive, online interpretation of Bayesian changepoint detection and provides a simple and exact method for calculating the posterior probability of the current run length. We have demonstrated this algorithm on three real-world data sets with different modelling requirements.

Additionally, this framework provides convenient delineation between the implementation of the changepoint algorithm and the implementation of the model. This modularity allows changepoint-detection code to use an object-oriented, “pluggable” type architecture.

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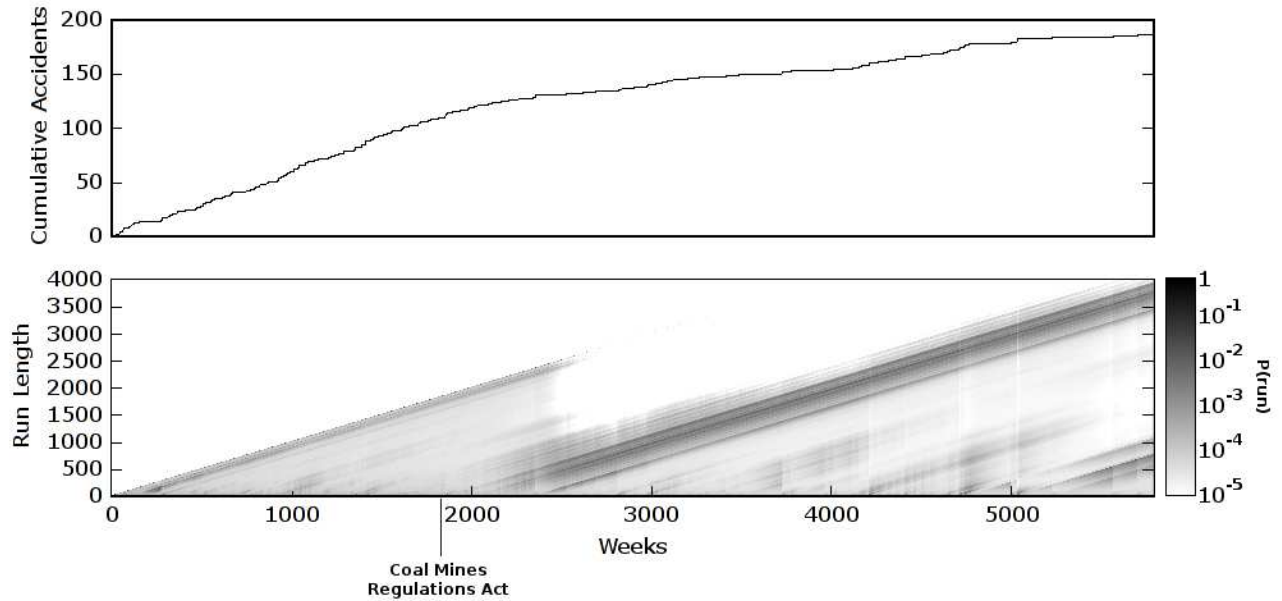


Figure 4: These data are the weekly occurrence of coal mine disasters that killed ten or more people between 1851 and 1962. The top plot is the cumulative number of accidents. The accident rate determines the local average slope of the plot. The introduction of the Coal Mines Regulations Act in 1887 is marked. The year 1887 corresponds to weeks 1868 to 1920 on this plot. The bottom plot shows the posterior probability of the current run length at each time step, $P(r_t | \mathbf{x}_{1:t})$.

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